FORMAL SMOOTHNESS DESCENDS ALONG ETALE MAPS OF SOURCE AND TARGET FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4553-4565 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4567-4575, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter lemmas that etale morphisms of algebraic spaces are formally etale (representable-property-formally-property), that formal smoothness composes with formal etaleness (composition-formally-smooth-etale-unramified) and the formal permanence criterion (lemma-formally-permanence); cited inputs with the chain written in the retained span.

Source reference. Original lemma 049T: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2486–2495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04G5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2666–2675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} U & \xrightarrow[\psi]{} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

be a commutative diagram of morphisms of algebraic spaces over S. If the vertical arrows are étale and f is formally smooth, then ψ is formally smooth.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. By Lemma 04AL the morphisms $U \to X$ and $V \to Y$ are formally étale. By Lemma 049T the composition $U \to Y$ is formally smooth. By Lemma 04G5 we see $\psi : U \to V$ is formally smooth. □